DATA SCIENCE LABORATORY MANUAL

Module I ? Data Science Fundamentals

Topic: Introduction to Data Science

OVERVIEW & LEARNING OBJECTIVES

Definition, domain scope, real-world applications, and core skill matrix of data scientists.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Introduction to Data Science in modern Data Science workflows.
- Detailed breakdown: Definition, domain scope, real-world applications, and core skill matrix of data scientists.
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.